WORKED SOLUTION

STEP III 2017, Question 13

Probability and Statistics • Expectation and Variance

STEP III 2017, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q13

The key idea

Write $X - x$ as $(X - \mu) + (\mu - x)$. The first piece has mean zero, so when we square and take expectations the cross term vanishes. That leaves the variance plus a constant, which is the same identity as "$\mathrm{E}(X^2) = \mathrm{Var}(X) + \mu^2$". For the uniform case, Y is a function of X that is not one-to-one, so the safest way to its density is through its cumulative distribution function.

Solution

The function V

For any fixed number x,

$$\begin{aligned}\mathrm{V}(x) = \mathrm{E}((X-x)^2) &= \mathrm{E}(((X-\mu)+(\mu-x))^2)\\ &= \mathrm{E}((X-\mu)^2) + 2(\mu-x)\,\mathrm{E}(X-\mu) + (\mu-x)^2\\ &= \sigma^2 + (x-\mu)^2,\end{aligned}$$

using linearity of expectation (here $\mu - x$ is a constant) and $\mathrm{E}(X - \mu) = 0$.

The mean of Y

Since $\mathrm{V}(x) = \sigma^2 + (x - \mu)^2$ holds for every real x, we may put $x = X$:

$$Y = \mathrm{V}(X) = \sigma^2 + (X-\mu)^2.$$

? **Ask yourself** Isn't $\mathrm{V}(X)$ just a number, since V was defined by an expectation?

No: V is an ordinary function of a real variable (a quadratic in x). The expectation was taken with x held fixed. Feeding the random variable X into that quadratic gives a new random variable. So

$$\mathrm{E}(Y) = \sigma^2 + \mathrm{E}((X-\mu)^2) = \sigma^2 + \sigma^2 = 2\sigma^2,$$

which is $(*)$.

The uniform case

For X uniform on $[0,1]$, $\mu = \frac{1}{2}$ and $\sigma^2 = \int_0^1 x^2\,\mathrm{d}x - \frac{1}{4} = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$. Hence

$$\mathrm{V}(x) = \tfrac{1}{12} + (x - \tfrac{1}{2})^2 = x^2 - x + \tfrac{1}{3}.$$

Distribution of Y. Now $Y = \frac{1}{12} + (X - \frac{1}{2})^2$. As X runs over $[0,1]$, $|X - \frac{1}{2}|$ runs over $[0, \frac{1}{2}]$, so Y takes values in $[\frac{1}{12}, \frac{1}{12} + \frac{1}{4}] = [\frac{1}{12}, \frac{1}{3}]$.

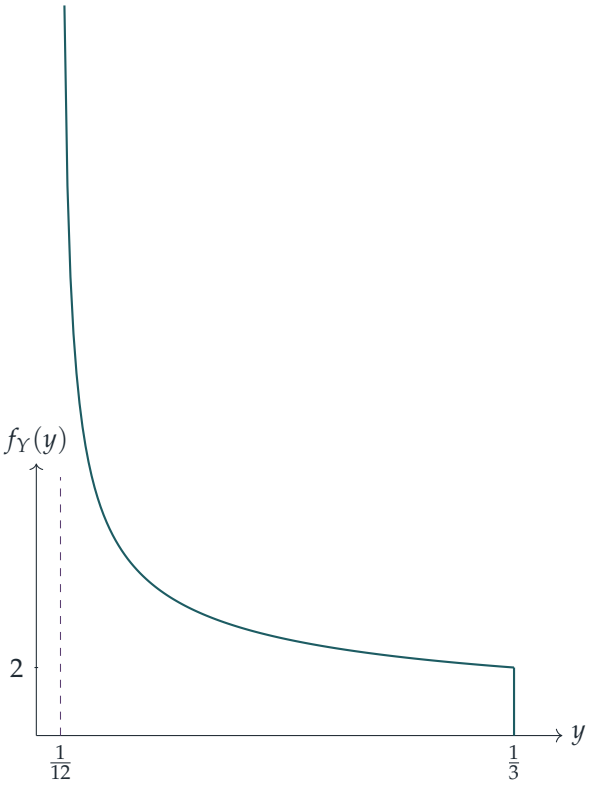
The map $X \mapsto Y$ is two-to-one (the points $\frac{1}{2} \pm h$ give the same Y), so we work with the distribution function. For $\frac{1}{12} \le y \le \frac{1}{3}$,

$$\begin{aligned} \mathrm{P}(Y \le y) &= \mathrm{P}((X - \tfrac{1}{2})^2 \le y - \tfrac{1}{12}) = \mathrm{P}(\tfrac{1}{2} - s \le X \le \tfrac{1}{2} + s), \quad \text{where } s = \sqrt{y - \tfrac{1}{12}}, \\ &= 2s = 2\sqrt{y - \tfrac{1}{12}}, \end{aligned}$$

because $0 \le s \le \frac{1}{2}$, so the interval lies inside $[0,1]$ and has length $2s$. (At $y = \frac{1}{3}$ this gives $2 \cdot \frac{1}{2} = 1$, as it should.) Differentiating,

$$f_Y(y) = \frac{1}{\sqrt{y - \frac{1}{12}}} \quad \text{for } \tfrac{1}{12} < y \le \tfrac{1}{3}, \qquad f_Y(y) = 0 \text{ otherwise.}$$

The density is unbounded near $y = \frac{1}{12}$, but it is still integrable: Y is most likely to be close to its minimum, because $(X - \frac{1}{2})^2$ is flat near $X = \frac{1}{2}$.



Checking $(*)$. Using the density, and substituting $t = y - \frac{1}{12}$ (so t runs from 0 to $\frac{1}{4}$),

$$\mathrm{E}(Y) = \int_{1/12}^{1/3} \frac{y}{\sqrt{y - \frac{1}{12}}}\,\mathrm{d}y = \int_0^{1/4} (t^{1/2} + \tfrac{1}{12}t^{-1/2})\,\mathrm{d}t = \left[\tfrac{2}{3}t^{3/2} + \tfrac{1}{6}t^{1/2}\right]_0^{1/4}$$
$$= \tfrac{2}{3}\cdot\tfrac{1}{8} + \tfrac{1}{6}\cdot\tfrac{1}{2} = \tfrac{1}{12} + \tfrac{1}{12} = \tfrac{1}{6}.$$

Since $2\sigma^2 = 2\cdot\frac{1}{12} = \frac{1}{6}$, this agrees with $(*)$.

Checks

- The density integrates to 1: $\int_0^{1/4} t^{-1/2}\,\mathrm{d}t = 2\cdot\frac{1}{2} = 1$.
- $\mathrm{V}(x)$ is smallest at $x = \mu$, with minimum value σ^2: the mean is the point that minimises expected squared distance. In the uniform case $\mathrm{V}(\frac{1}{2}) = \frac{1}{12}$, the bottom of the range of Y.
- A computer simulation with 400 000 uniform values gave a sample mean of Y of about 0.1667, matching $\frac{1}{6}$.

Common slips

- Treating $\mathrm{V}(X)$ as the constant σ^2, which gives $\mathrm{E}(Y) = \sigma^2$.
- Using the one-to-one change-of-variable formula $f_Y = f_X \left|\mathrm{d}x/\mathrm{d}y\right|$ with only one branch, which gives half the correct density. Both $X = \frac{1}{2} + s$ and $X = \frac{1}{2} - s$ contribute.

Beyond STEP

$\mathrm{V}(x) = \sigma^2 + (x - \mu)^2$ is the simplest bias–variance decomposition. If you estimate a quantity by x, the mean squared error splits into the spread of the data plus the squared bias. The result $\mathrm{E}(Y) = 2\sigma^2$ says that the expected squared distance between two independent copies of X is $2\sigma^2$. Statisticians use this fact again and again, for example when explaining why the sample variance divides by $n - 1$. Blitzstein and Hwang, *Introduction to Probability*, covers both ideas, and also finding the density of a non-monotonic function of a random variable.

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